

KHORRAMABAD, Iran, Islamic Republic ofWMO: **407820**Lat: **33.433N**Lon: **48.283E**Elev: **3766**StdP: **12.80**Time Zone: **3.50 (IRN)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.7	27.6	8.6	9.8	49.2	12.3	11.8	50.1	19.4	46.1	16.8	45.2	2.4	120	0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	33.0	105.9	63.1	104.0	62.7	102.1	62.4	66.3	98.5	64.7	98.1	63.6	97.2	8.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.8	79.1	69.6	54.0	71.4	65.5	52.5	67.5	64.8	33.0	98.7	31.7	97.8	30.8	97.6	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	15.7	13.5	DB	19.5	109.8	5.3	1.7	15.6	111.0	12.5	112.0	9.6	113.0	5.7	114.2	(4)
(5)			WB	17.7	67.4	5.0	2.8	14.1	69.4	11.2	71.0	8.4	72.5	4.7	74.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.2	41.0	44.9	52.0	58.8	69.4	80.3	86.2	85.5	76.3	65.8	52.3	44.5	(6)	
	DBStd	16.52	5.06	5.68	5.83	5.40	5.23	3.91	3.63	3.58	4.54	5.30	5.78	4.58	(7)	
	HDD50	706	281	157	43	3	0	0	0	0	0	0	41	181	(8)	
	HDD65	3003	745	563	405	194	21	0	0	0	0	55	383	637	(9)	
	CDD50	5522	2	14	106	268	601	908	1124	1101	788	491	109	10	(10)	
	CDD65	2341	0	0	2	9	157	458	659	636	338	81	1	0	(11)	
	CDH74	36866	0	0	50	243	2452	7231	10418	9977	5091	1389	15	0	(12)	
	CDH80	22933	0	0	5	42	1066	4625	7046	6722	2960	467	0	0	(13)	
(14) Wind	WSAvg	5.2	4.6	5.4	5.8	5.5	5.9	5.9	5.7	5.6	5.3	4.9	4.0	4.0	(14)	
(15) Precipitation	PrecAvg	22.2	3.2	3.0	4.0	3.6	0.7	0.1	0.0	0.0	0.1	1.2	2.7	3.6	(15)	
	PrecMax	33.3	5.2	8.6	8.7	8.3	2.4	0.5	0.1	0.1	0.6	5.2	11.5	8.2	(16)	
	PrecMin	13.0	0.4	0.6	0.6	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	(17)	
	PrecStd	5.0	1.3	1.9	2.1	2.0	0.7	0.1	0.0	0.0	0.2	1.4	2.2	2.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.0	67.1	80.3	84.6	95.4	105.4	109.2	107.8	102.0	91.2	76.1	67.8	(19)	
		MCWB	45.3	48.7	53.2	58.6	59.8	63.5	64.0	63.7	61.7	57.7	52.6	46.9	(20)	
	2%	DB	57.4	62.9	73.4	79.2	92.5	102.3	106.4	105.8	98.8	88.0	71.9	61.2	(21)	
		MCWB	44.1	46.3	50.3	56.6	59.2	61.9	63.3	62.7	60.7	56.8	51.5	46.5	(22)	
	5%	DB	55.0	60.4	69.4	76.6	89.4	100.3	104.4	104.0	96.7	85.6	68.2	57.8	(23)	
		MCWB	43.7	45.4	49.5	55.2	58.2	61.3	63.0	62.6	59.6	55.9	50.4	45.4	(24)	
	10%	DB	51.8	57.1	65.8	73.3	86.1	98.3	102.3	102.1	93.7	82.3	64.4	54.8	(25)	
		MCWB	41.8	44.2	48.3	54.1	57.4	60.5	62.8	62.3	58.5	54.7	49.5	44.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.1	52.0	55.2	61.6	63.9	65.8	72.6	68.0	65.3	60.8	57.2	51.6	(27)	
		MCDB	54.6	60.4	68.6	76.3	85.5	96.6	97.5	99.5	95.9	81.7	66.1	58.8	(28)	
	2%	WB	46.9	49.1	53.4	59.2	62.1	63.9	67.2	65.6	62.7	59.0	54.9	49.1	(29)	
		MCDB	53.1	58.1	68.2	74.3	83.3	96.6	98.2	100.2	92.1	78.9	63.5	56.0	(30)	
	5%	WB	45.3	47.0	51.6	57.5	60.8	62.8	65.2	64.3	60.8	57.4	53.3	47.5	(31)	
		MCDB	51.8	57.2	64.6	70.9	80.6	96.0	98.7	99.6	91.5	76.7	63.1	54.0	(32)	
	10%	WB	43.4	45.5	49.8	56.0	59.5	61.5	63.8	63.1	59.4	56.2	51.8	46.0	(33)	
		MCDB	49.7	55.5	62.3	68.5	79.4	94.7	98.4	98.7	90.8	75.6	61.1	52.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	20.1	21.8	24.3	24.9	29.5	34.9	33.0	34.1	35.4	29.8	23.5	21.2	(35)	
		MCDBR	25.3	26.9	30.2	31.1	34.6	37.2	34.7	35.1	36.6	34.0	28.9	26.8	(36)	
		MCWBR	14.2	13.9	13.5	12.8	11.6	13.1	11.7	12.1	13.7	14.1	13.5	14.1	(37)	
	5% WB	MCDBR	19.9	22.3	25.2	25.4	29.3	34.6	31.8	33.1	34.0	28.0	20.6	19.6	(38)	
		MCWBR	12.0	12.2	12.5	11.6	11.1	12.6	12.0	12.0	13.2	12.1	10.4	11.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.349	0.401	0.470	0.572	0.547	0.488	0.548	0.501	0.424	0.462	0.394	0.350	(40)	
	taud		2.200	2.045	1.868	1.687	1.757	1.878	1.755	1.856	2.027	1.924	2.093	2.217	(41)	
	Ebn at Noon		268	264	254	234	242	256	240	249	265	239	247	259	(42)	
	Edh at Noon		36	47	61	76	72	64	72	64	51	53	40	34	(43)	
(44) All-Sky Solar Radiation	RadAvg		907	1182	1569	1831	2223	2642	2476	2315	2025	1470	1039	839	(44)	
	RadStd		53	92	137	163	154	87	88	84	71	108	97	96	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	+1.12	-2.32	-1.89	N/A	N/A	N/A	N/A

Nomenclature: See separate page